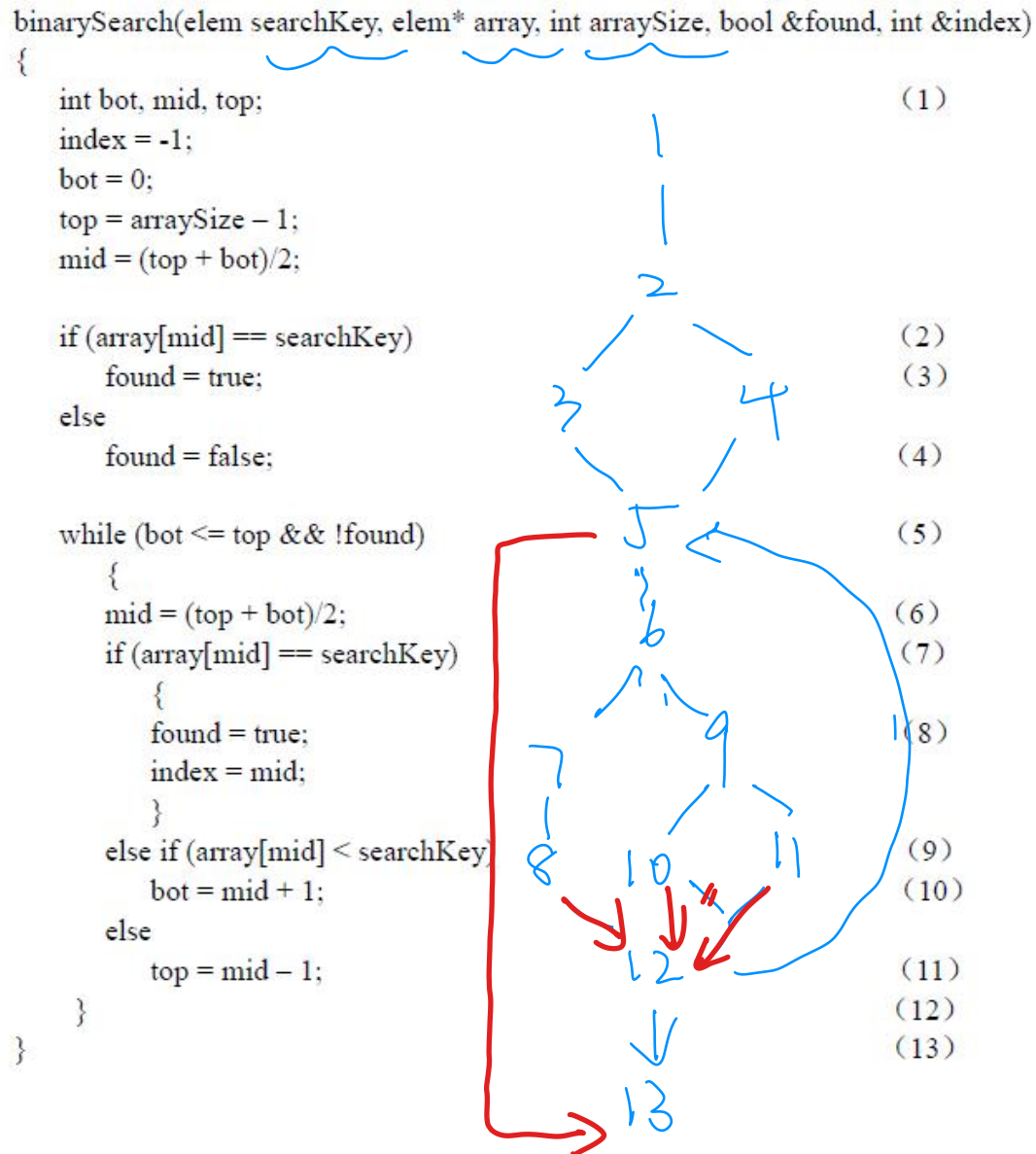


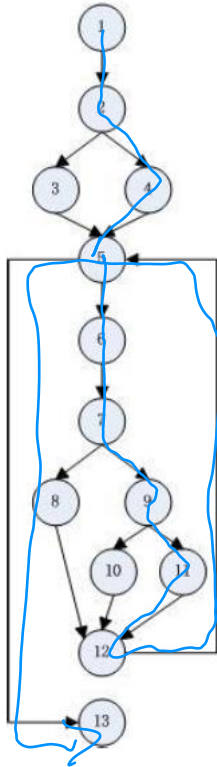
Tutorial – Week 12**White-box Testing**

The above is a binary search function written in C language. Answer the following questions base on the code.

Question 1

Draw a control flow graph corresponding to the preceding function.

Answer



1-2-3-5-13

1-2-4-5-13

1-2-3-5-6-7-8-12-5-13

Question 2

For the preceding function, derive a set of test cases that will cover branch testing.

分支测试就是把代码判断两种结果跑一遍

Answer

	Path	Test Case
P1	1-2-3-5-13	arraysize = 3, searchkey = b, array = abc
P2	1-2-4-5-13	arraysize = 0, searchkey = a
P3	1-2-4-5-6-7-8-12-5-13	impossible
P4	1-2-4-5-6-7-9-10-12-5-6-7-8-12-5-13	arraysize = 3, searchkey = a, array = abc
P5	1-2-4-5-6-7-9-11-12-5-6-7-8-12-5-13	arraysize = 3, searchkey = c, array = abc

1-2-3-5-6-7-9-11-12-5-13

Question 3

Design additional test cases that will cover multiple-condition testing.

Answer

According to *while (bot <= top && !found)*, Multiple-condition testing is as follows

bot <= top	! found	Test Case
T	T	arraysize = 3, searchkey = a, array = abc
T	F	arraysize = 3, searchkey = b, array = abc
F	T	arraysize = 3, searchkey = d, array = abc
F	F	impossible

4种

多条件测试：子条件组成复合判断
 断把每个子条件T/F排列。

Question 4

Determine the cyclomatic complexity of the preceding function.

Answer

$$V(G) = e - n + 2p = 16 - 13 + 2 = 5$$

Question 5

Derive a set of test cases that cover McCabe's Basis Paths.

Answer

	Path	Test Case
P1	1-2-4-5-6-7-9-10-12-5-6-7-9-11-12-5-6-7-8-12-5-13	arraysize = 7, searchkey = e, array = abcdefg
P2: Flip at 2	1-2-3-5-6-7-9-10-12-5-6-7-9-11-12-5-6-7-8-12-5-13	impossible
P3: Flip at 5	1-2-4-5-13	arraysize = 0, searchkey = a
P4: Flip at 7	1-2-4-5-6-7-8-12-5-6-7-9-11-12-5-6-7-8-12-5-13	impossible
P5: Flip at 9	1-2-4-5-6-7-9-11-12-5-6-7-9-11-12-5-6-7-8-12-5-13	arraysize = 7, searchkey = a, array = abcdefg

最多分支